

Performance Metric

↑
Classification
↑

Regression

1400
100
(0,1)

↑
(y_{pred})

80
1000 → 80%
↑
80%
↑

But =

80%

Acc = $\frac{TP}{TP+FP}$

200
O(x,y) → Train (x_{train}, y_{train})
500
Dev (x_{test}, y_{test})

(0,1)
200
1800

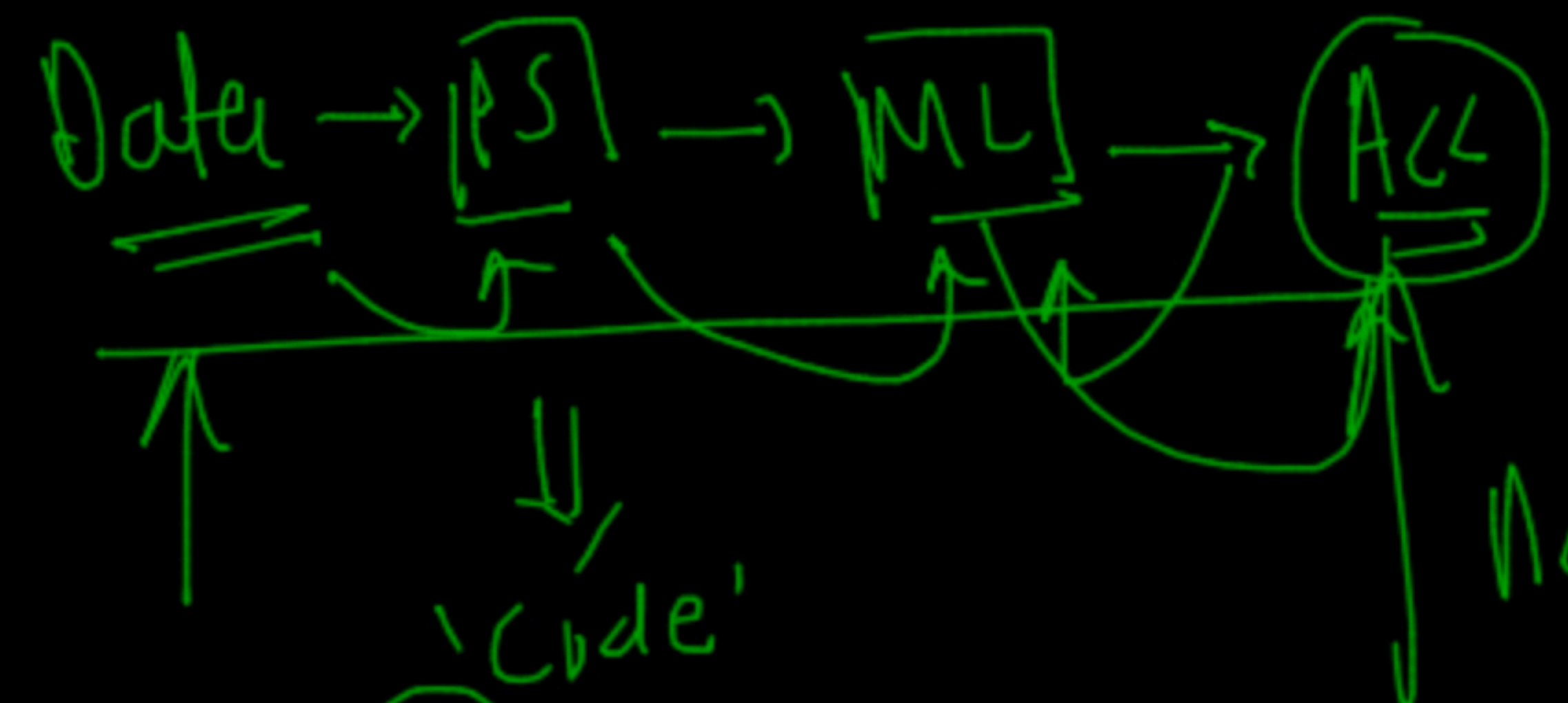
Imbalanced

0 → 50
1 → 450

X	Y
x ₁	1
x ₂	1
...	...
x _n	1

400 → 50%
BND

~ {x, y} = 50%



imbyla

DS (lead)

KNN
 $ML() \rightarrow (1, 1, 0)$

$knn = KNN()$
 $knn.bit()$
 $knn.predict()$
 $(1, 1, 0)$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(prob-score)

X	Y	M1	M2	Y1	Y2
x1	1 ✓	0.9	0.6	1 ✓	1 ✓
x2	1 ✓	0.8	0.65	1 ✓	1 ✓
x3	0 ✓	0.1	0.15	0 ✓	0 ✓
x4	0 ✓	0.15	0.48	0 ✓	0 ✓

$\begin{pmatrix} 4 \\ 4 \end{pmatrix}$ $\rightarrow 100\%$

if prob score > 0.5
— 1

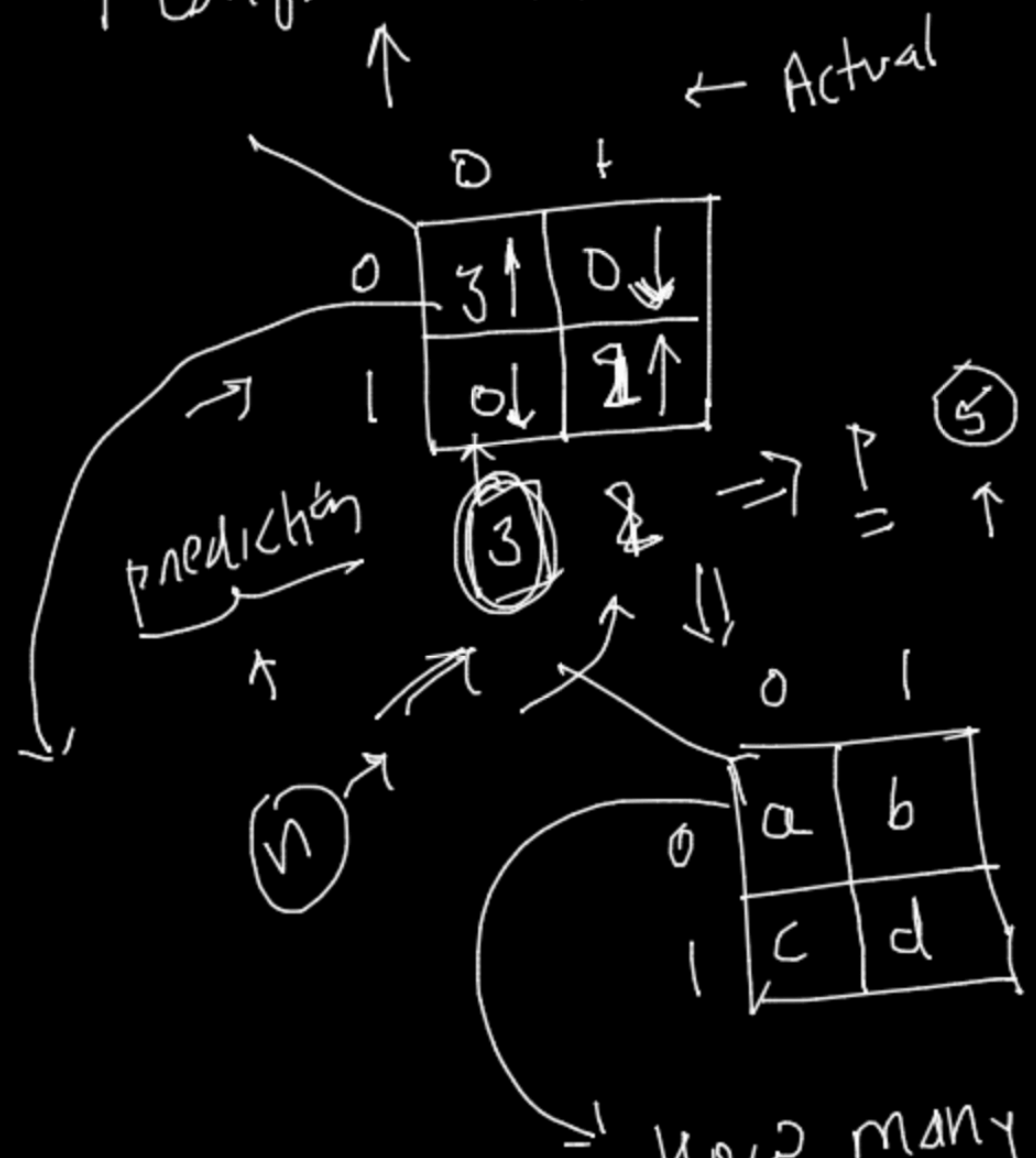
$\begin{bmatrix} 4 \\ \rightarrow \\ 4 \end{bmatrix}$

$\rightarrow 100\%$

$knn.predict \rightarrow prob(p)$
(store) $\downarrow$
 $\begin{bmatrix} 0 \rightarrow \\ 1 \end{bmatrix}$

1) ACC (Classification)

2) Confusion Matrix



X	Y	Z
	0	0
	0	0
	0	0
	1	1
	1	1

	0	1	2
0	↑	↓	↓
1	↓	↑	↓
2	↓	↑	↑

How many data points which have predicted value as '0' and actual value is also '0' →

Negative (Ans)

	0	1
0	TN	FN
1	FP	TP

(TN ↑ TP ↑)

(FP ↓ FN ↓)

$N + P = n$

↑ total data points



one left → what is predicted
correct label

(i) TPR (True +ve Rate) =

$$\frac{\#TP}{P}$$

(ii) TNR = $\frac{\#TN}{N}$

(iii) FPR = $\frac{FP}{P}$

(iv) FNR = $\frac{FN}{N}$

good model
[TPR and TNR ↑]

eg.

	0	1
0	850	6
1	50	94

i) $TPR = \frac{94}{100} \rightarrow 94\%$

ii) $TNR = \frac{850}{950} \rightarrow \uparrow$

	0	1
0	850	6
1	50	94

↓
500
=

iii) $FNR = \frac{6}{100} \rightarrow 6\%$

iv) $FPR = \frac{50}{950}$

1000
↑
qno (+ve)
100 (-ve)

	0	1
0	0	50
1	100	850

model ↓

TNR → 0%

↑
worst

x	y	z
	✓	✓

	0	1
0	0	
1		0

Code → sklearn.metrics → confusion

← Accuracy ← (TN)

Σ 16
y=0 & y=0 → (only -)

⇒ Precision, Recall and F1-score

$$\boxed{TPR = \frac{TP}{P}}$$



$$\frac{TP}{TP + FP}$$

TP

Y	X
0	
0	
0	
0	
1	1
1	1
1	6
1	9

of all the points the model
declared/predicted to be +ve,
what %age of them are
actually +ve

$$\frac{4}{7} \rightarrow \frac{50}{100}$$

0 → ✓
1 → ✓

$$\underline{\underline{F1\text{-}score}} = 2 * \left(\frac{Pr * Recall}{Pr + Recall} \right) \leftarrow \underline{\underline{formula}}$$

good model

$$\underline{\underline{F1\text{-}score}} \uparrow \rightarrow \textcircled{1}$$

$$\begin{matrix} 0 & \text{---} & 1 \\ \uparrow & & \\ model & & \end{matrix}$$

$\begin{cases} Pr \\ R \end{cases}$